

CSS Research Project

1. `if()`:

Year of Introduction / Standardization: 2026

Syntax: `if(condition: value; else: value)`

Purpose / Use Case: Allows the use of conditionals in CSS directly thereby reducing dependencies on JavaScript.

Small Working Code Example:

Browser Support Information: Chrome 137+, Edge 137+, Safari Technology Preview, Firefox

Reference Source (MDN / W3C / Official Drafts): MDN

2. `display: grid-lanes`

Year of Introduction / Standardization: 2025

Syntax: `display: grid-lanes;`

Purpose / Use Case: Allows the use of pinterest-style grid. Where vertical spaces are filled seamlessly.

Small Working Code Example:

```
.gallery {  
  
  display: grid-lanes;  
  
  grid-template-rows: masonry;  
  
  gap: 10px;  
  
}
```

Browser Support Information: Safari TP 234+, Chrome 140+

Reference Source (MDN / W3C / Official Drafts): W3C

3. anchor()

Year of Introduction / Standardization: 2025-2026

Syntax: top: anchor(left);

Purpose / Use Case: Positions elements relative to other anchor elements.

Small Working Code Example:

```
.tooltip {  
  
    position: absolute;  
  
    inset-block-end: anchor(top);  
  
    inset-inline-start: anchor(end);  
  
}
```

Browser Support Information: Chrome 137+, Safari 19+, Firefox

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

4. view-timeline

Year of Introduction / Standardization: 2026

Syntax: view-timeline: --name inline;

Purpose / Use Case: Allows scroll driven animations.

Small Working Code Example:

```
.box {  
  
    view-timeline: --scroll inline;  
  
    animation-timeline: --scroll;  
  
}
```

Browser Support Information: Chrome 137+, Safari 19+, Firefox

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

5. sibling-index()

Year of Introduction / Standardization: 2025

Syntax: sibling-index()

Purpose / Use Case: Returns 1-based indexes of an element among its siblings.

Small Working Code Example:

```
li {  
  
  width: calc(sibling-index() * 50px);  
  
}
```

Browser Support Information: Chrome, Safari,

Reference Source (MDN / W3C / Official Drafts): MDN

6. sibling-count()

Year of Introduction / Standardization: 2025

Syntax: sibling-count()

Purpose / Use Case: Returns total number of siblings.

Small Working Code Example:

```
.item {  
  
  flex: calc(1 / sibling-count());  
  
}
```

Browser Support Information: Chrome, Safari,

Reference Source (MDN / W3C / Official Drafts): MDN

7. @scope

Year of Introduction / Standardization: 2025

Syntax:

```
@scope (.root) {  
  
  ...  
  
}
```

Purpose / Use Case: limits style to specific DOM subtree

Small Working Code Example:

```
@scope (.card) {  
  
  .title {  
  
    color: blue;  
  
  }  
  
}
```

Browser Support Information: Chrome, Safari, Firefox

Reference Source (MDN / W3C / Official Drafts): MDN

8. corner-shape

Year of Introduction / Standardization: 2025

Syntax: corner-shape: bevel;

Purpose / Use Case: Allows using different shapes for corners

Small Working Code Example:

```
.btn {  
  
  border-radius: 15px;
```

```
corner-shape: bevel;  
}
```

Browser Support Information: Chrome, Safari

Reference Source (MDN / W3C / Official Drafts): MDN

9. Container-type: scroll-state

Year of Introduction / Standardization: 2025

Syntax: container-type: scroll-state;

Purpose / Use Case: Allows querying scroll state of container.

Small Working Code Example:

```
.container {  
    container-type: scroll-state;  
}
```

Browser Support Information: Chrome

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

10. position-area

Year of Introduction / Standardization: 2025

Syntax: position-area: bottom center;

Purpose / Use Case: Grid-based positioning relative to anchor

Small Working Code Example:

```
.popover {  
    position-area: bottom center;
```

}

Browser Support Information: Chrome

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

11. scroll-timeline

Year of Introduction / Standardization: 2026

Syntax: scroll-timeline: --name block;

Purpose / Use Case: Defines the scroll based animations

Small Working Code Example:

```
.progress {  
  
  scroll-timeline: --bar inline;  
  
  animation-timeline: --bar;  
  
}
```

Browser Support Information: Chrome

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

12. timeline-trigger-name

Year of Introduction / Standardization: 2026

Syntax: timeline-trigger-name: --enter;

Purpose / Use Case: Triggers animations tied to timelines

Browser Support Information: Chrome

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

13. flow-tolerance

Year of Introduction / Standardization: 2025

Syntax: flow-tolerance: normal |percentage | length;

Purpose / Use Case: Controls the tolerance allowed when placing items in grid-lanes layouts. Helps in managing gaps and layout.

Small Working Code Example:

```
.masonry {  
  
  display: grid-lanes;  
  
  grid-template-rows: masonry;  
  
  flow-tolerance: 10%;  
  
}
```

Browser Support Information: Chrome, Safari

Reference Source (MDN / W3C / Official Drafts): W3C

14. item-flow

Year of Introduction / Standardization: 2025

Syntax: item-flow: row | column | dense | pack;

Purpose / Use Case: Controls how items are placed and packed inside complex grid layouts

Small Working Code Example:

```
.grid {  
  
  display: grid-lanes;  
  
  item-flow: column dense;  
  
}
```

Browser Support Information: Chrome, Safari

Reference Source (MDN / W3C / Official Drafts): W3C

15. anchor-center

Year of Introduction / Standardization: 2025

Syntax:

align-self: anchor-center;

justify-self: anchor-center;

Purpose / Use Case: Centers a position element relative to its anchor element without math calculations.

Small Working Code Example:

```
.modal {  
  
  position: absolute;  
  
  align-self: anchor-center;  
  
  justify-self: anchor-center;  
  
}
```

Browser Support Information: Chrome, Safari, Firefox

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

16. shape()

Year of Introduction / Standardization: 2026

Syntax: clip-path: shape(<shape-function>);

Purpose / Use Case: Extends CSS shape functions for advanced clippings

Small Working Code Example:

```
.tag {
```

```
clip-path: shape(polygon(0% 0%, 100% 0%, 90% 100%, 0% 100%));  
}
```

Browser Support Information: Chrome, Safari

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

17. text-wrap: balance;

Year of Introduction / Standardization: 2026

Syntax: text-wrap: balance;

Purpose / Use Case: Balances text across lines for improved readability

Small Working Code Example:

```
h1 {  
  
    text-wrap: balance;  
  
    max-width: 400px;  
  
}
```

Browser Support Information: Chrome, Safari, Firefox

Reference Source (MDN / W3C / Official Drafts): MDN

18. light-dark()

Year of Introduction / Standardization: 2026

Syntax: light-dark(<light-color>, <dark-color>)

Purpose / Use Case: Automatically switches color values based on user's light/dark theme

Small Working Code Example:

```
body {
```

```
background-color: light-dark(white, black);  
  
color: light-dark(black, white);  
  
}
```

Browser Support Information: Chrome, Safari, Firefox

Reference Source (MDN / W3C / Official Drafts): Chrome for developers

19. color-mix()

Year of Introduction / Standardization: 2026

Syntax: color-mix(in color-space, color1 percentage, color2)

Purpose / Use Case: Dynamically mixes two colours in a chosen place

Small Working Code Example:

```
.box {  
  
background-color: color-mix(in oklch, blue 60%, white);  
  
}
```

Browser Support Information: Chrome, Safari, Firefox

Reference Source (MDN / W3C / Official Drafts): MDN

20. math-depth

Year of Introduction / Standardization: 2026

Syntax: math-depth: auto | integer;

Purpose / Use Case: Controls visuals depth of nested MathML elements

Small Working Code Example:

```
math {
```

math-depth: auto;

}

Browser Support Information: Chrome, Safari, Firefox

Reference Source (MDN / W3C / Official Drafts): MDN